

作业可中英 可电子可手写

补充习题后有历年试卷和中英文对照

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Office Hours: Tues 10:00~12:00, 15:30~17:00
(冲突可发邮件预约)

2.3 不作要求 (极限的严格定义)

6.5 ——— 课程大纲上红色不作要求

教材: Thomas' Calculus (13th edition) (托马斯微积分).

辅导教材: 同济 (2007). 不建议 概念有差异
体系不同 (No credits using
老师会补例题 一期中 各分达)

刷题适可而止

11.2
期中: 第八周 (无补考)

出勤	5%	
作业	25%	(教材 15% 非教材 10%)
期中	30%	
期末	40%	

换班要带着作业走

作业没有答案

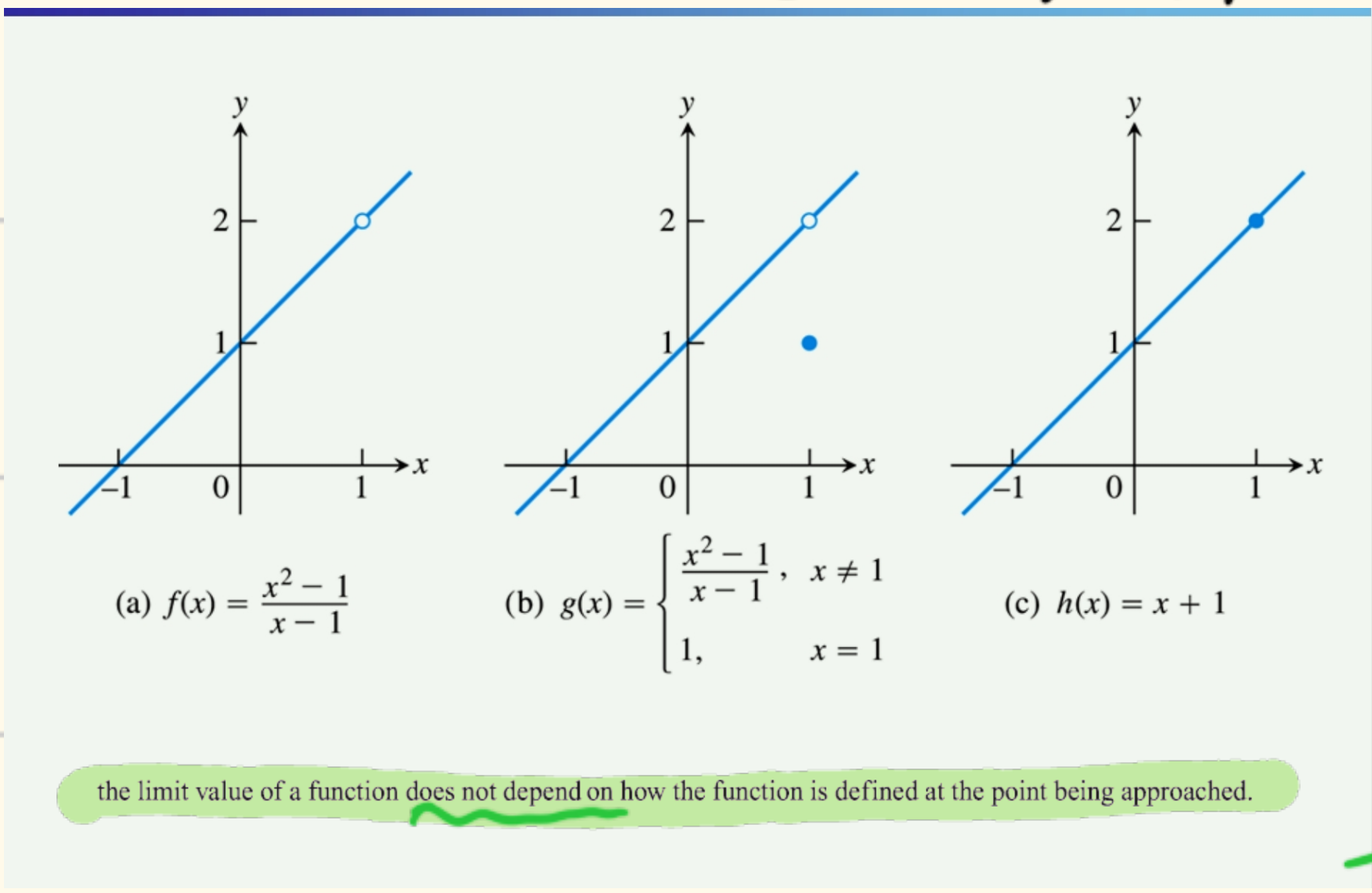
<< Mental Abuse to Health >>

<< My Approach to Happiness >>

rate of change } average —

$\lim_{x \rightarrow 0} f(x) = L$

$\lim_{x \rightarrow 0}$ 当...时
 $f(x)$ 关注对象
 $= L$ 趋于; go to, approach, ...



极限均为2.

与该点无关 (nearby)

极限计算:

$f(x) = k \quad \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} k = k$
 $f(x) = x \quad \lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} x = c$

(左极限和右极限)
 不相同视为无极限
 左右都有且相同
 $y = \begin{cases} 0, & x \leq 0 \\ \sin \frac{1}{x}, & x > 0 \end{cases}$
 在 0^+ 附近变化过快

Limit Laws: $\lim_{x \rightarrow c} f(x) = L \quad \lim_{x \rightarrow c} g(x) = M$

$\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$ the limit of the sum = the sum of the limits
 $\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$

THEOREM 1—Limit Laws

If L, M, c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M,$$
 then

1. Sum Rule:	$\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$
2. Difference Rule:	$\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$
3. Constant Multiple Rule:	$\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$
4. Product Rule:	$\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$
5. Quotient Rule:	$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$ Check!
6. Power Rule:	$\lim_{x \rightarrow c} [f(x)]^n = L^n, n \text{ a positive integer}$
7. Root Rule:	$\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, n \text{ a positive integer}$

(If n is even, we assume that $\lim_{x \rightarrow c} f(x) = L > 0$.)

$$P(x) = A_n x^n + A_{n-1} x^{n-1} + \dots + A_0$$

$$\lim_{x \rightarrow c} P(x) = \lim_{x \rightarrow c} (A_n x^n) + \dots + \lim_{x \rightarrow c} A_0$$

$$= A_n \lim_{x \rightarrow c} (x^n) + \dots + \lim_{x \rightarrow c} A_0$$

$$= A_n (\lim_{x \rightarrow c} x)^n + \dots + \lim_{x \rightarrow c} A_0$$

$$= A_n c^n + A_{n-1} c^{n-1} + \dots + A_0$$

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}$$

$$\frac{1+1-2}{1-1} : \frac{0}{0} \text{ 型} = \sum_{i=0}^n A_i c^i$$

消去0因子(因式分解)

$$= \lim_{x \rightarrow 1} \frac{\cancel{(x-1)}(x+2)}{x\cancel{(x-1)}}$$

(if $x \neq 1$) 由于 x 趋于 1 所以 $x \neq 1$.
可这样约分

$$= \lim_{x \rightarrow 1} (x+2) = 3$$

计算极限

① 交换运算顺序 (前提: 各部分均可计算)

$\frac{L}{M}$ 中 $M \neq 0$

M, L 均为 0: 消 0 因子

② Sandwich Theorem

③ 重要极限

$$\lim_{x \rightarrow 1} \frac{(1-\sqrt{x})(1-\sqrt[3]{x})}{(1-x^2)^2} = \lim_{x \rightarrow 1} \frac{(1-\sqrt{x})(1-\sqrt[3]{x})}{(1-x)^2(1+x)^2}$$

非 0 常数因子

$$= \frac{1}{4} \lim_{x \rightarrow 1} \frac{\cancel{(1-\sqrt{x})} \cancel{(1-\sqrt[3]{x})}}{\cancel{(1-\sqrt{x})} (1+\sqrt{x}) (1-x)}$$

$$= \frac{1}{4} \lim_{x \rightarrow 1} \frac{1}{(1+\sqrt{x})(1+\sqrt[3]{x}+\sqrt{x^2})}$$

$$= \frac{1}{4} \times \frac{1}{2 \times 3} = \frac{1}{24}$$

$$= \cancel{(1-\sqrt{x})} (1+\sqrt{x}+\sqrt{x^2})$$

有 + 无 = 无

无 + 无 = 不定

有 + 有 = 有

有 $\times$ 无 = 不定

无 $\times$ 无 = 不定

$\uparrow$ $\begin{cases} L=0 \\ L \neq 0 \end{cases}$ (1/a) e.g. $\lim_{x \rightarrow 0} (x \cdot \frac{1}{x})$

Sandwich Theorem:

Suppose that $g(x) \leq f(x) \leq h(x)$ (c处可不符合)

Suppose also that $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$,

Then $\lim_{x \rightarrow c} f(x) = L$.

$$\begin{aligned} |\sin x| &\leq 1 \\ |\cos x| &\leq 1 \end{aligned}$$

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x}: \quad -|x| \leq x \sin \frac{1}{x} \leq |x|$$

$$\lim_{x \rightarrow 0} |f(x)| = 0. \quad \text{Since } -|f(x)| \leq f(x) \leq |f(x)|,$$

$|f(x)|$ limit 0 as $x \rightarrow c$.

$$\text{So } \lim_{x \rightarrow 0} f(x) = 0.$$

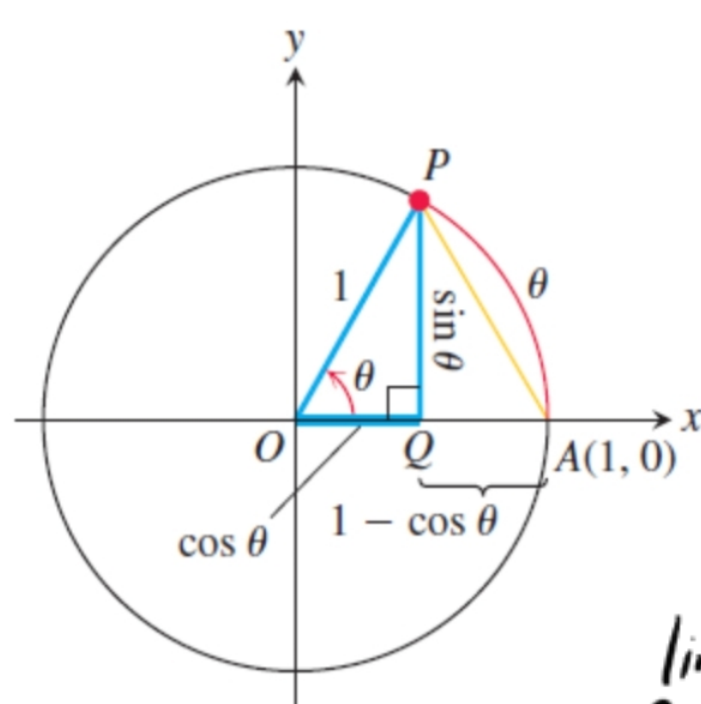
保号性: $f(x) \leq g(x)$ 同 c 无 $\neq$

$$\text{则 } \lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} g(x).$$

$$\begin{aligned} f(x) < g(x): \\ \lim_{x \rightarrow 0} f(x) &\leq \lim_{x \rightarrow 0} g(x). \\ \text{e.g. } 0 &\leq \frac{1}{n}. \end{aligned}$$

For any angle θ measured in radians, the sine and cosine functions satisfy

$$-|\theta| \leq \sin \theta \leq |\theta| \quad \text{and} \quad -|\theta| \leq 1 - \cos \theta \leq |\theta|.$$



(a) $\lim_{\theta \rightarrow 0} \sin \theta = 0$

(b) $\lim_{\theta \rightarrow 0} \cos \theta = 1$

$$\begin{aligned} (-|\theta| \leq \sin \theta \leq |\theta|) \\ \lim_{\theta \rightarrow 0} (1 - \cos \theta) = 0 = 1 - \lim_{\theta \rightarrow 0} \cos \theta. \end{aligned}$$

$$\begin{aligned} \lim_{\theta \rightarrow \theta_0} \sin \theta &= \lim_{\theta \rightarrow \theta_0} \sin(\underbrace{\theta - \theta_0}_{(x)} + \theta_0) \\ &= \lim_{\theta \rightarrow \theta_0} \sin \theta_0 \end{aligned}$$

The precise definition: 距离: 绝对值

无限接近: $\forall \varepsilon > 0, \delta > 0, 0 <$

$$0 < |x - x_0| < \varepsilon \Rightarrow |f(x) - L| < \delta$$

定义: ① 不等式不相解 ② 无穷小极限

$$\lim_{x \rightarrow c} f(x) = L \Leftrightarrow \lim_{x \rightarrow c^+} f(x) = L \text{ and } \lim_{x \rightarrow c^-} f(x) = L.$$

分左右看: ① 分板 ② $\frac{1}{x}$ (走势突变)

One-sided Limits (单边极限): $c - \delta < x < c$ (左)
 $c < x < c + \delta$ (右)

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0: \delta > 0 \quad 0 < x < \delta \quad |\sqrt{x} - 0| < \delta$$

Significant Limit: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

$$\lim_{\square \rightarrow 0} \frac{\sin \square}{\square} = 1$$

$$\lim_{x \rightarrow \infty} x \sin \frac{1}{x} = 1$$

$$\frac{1}{x} \rightarrow 0. \left(\frac{\sin \frac{1}{x}}{\frac{1}{x}} \right)$$

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{\tan t \sec 2t}{3t} &= \lim_{t \rightarrow 0} \frac{1}{3} \cdot \frac{1}{t} \cdot \frac{\sin t}{\cos t} \cdot \frac{1}{\cos 2t} \quad \text{① } \frac{\sin t}{t} \\ &\quad \text{② } 0/0 \text{ 子} \\ &= \frac{1}{3} \times 1 \times \frac{1}{1} = \frac{1}{3}. \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin(x^3)} = \lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x} - \sin x}{\sin(x^3)} = \lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{\cos x \sin(x^3)}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x \cdot \sin^2 \frac{x}{2}}{\cos x \sin x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin x}{x} \cdot x \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \right)^2 \cdot \left(\frac{x}{2} \right)^2}{\sin(x^3)} =$$

$$\sin x = \frac{\sin x}{x} \cdot x = x \quad (x \rightarrow 0)$$

因子时可代 (仅参与乘除 加减不换)

再次强调: $x \rightarrow 0$. 先看当 $x \rightarrow 0$ 时

连续:

函数与极限可以交换顺序

$$\lim_{x \rightarrow c} g(f(x)) = g(\lim_{x \rightarrow c} f(x)) = g(f(c))$$

在 $x=c$ 处连续: $\lim_{x \rightarrow c} f(x) = f(c)$

在 (a,b) 上连续: 中间两侧连续 端点仅一边连续

不连续: removable; jump discontinuity;

infinite discontinuity;

oscillating discontinuity.

$y = \frac{1}{x}$: is continuous over its natural domain

is discontinuous on any interval containing

初等函数: 有表达式即为连续函数

$x=0$.

$$\lim_{x \rightarrow c} (f+g)(x) = \lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$$

$$= f(c) + g(c) = (f+g)(c)$$

连续函数 algebraic combinations 仍为连续函数.

(能代直接代)

绝对值函数: 是连续函数. (脱号讨论: $f(x) = \pm x$ 连续.

$$x=0: \text{证 } \lim_{x \rightarrow 0^+} = \lim_{x \rightarrow 0^-} = 0 = f(0)$$

三角函数: $\checkmark$ (只要有定义就是连续函数)

$$(g \circ f)(x) = g(f(x))$$

f continuous at c , g continuous at $f(c) \Rightarrow g \circ f$ continuous at c .

$$g \text{ --- at } b, \lim_{x \rightarrow c} f(x) = b \Rightarrow \lim_{x \rightarrow c} g(f(x)) = g(b) = g(\lim_{x \rightarrow c} f(x))$$

3. 证明连续:

$$\lim_{x \rightarrow c} f(x) = b \quad \lim_{x \rightarrow b} g(x) = L \not\Rightarrow \lim_{x \rightarrow c} g(f(x)) = L \quad (g(x) \text{ 不一定连续})$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{\sqrt{1+x} - 1} \quad \text{换元: } u = \sqrt{1+x} \quad (\text{复合函数}) \quad g(\lim_{x \rightarrow c} f(x)) = g(b) \neq \lim_{x \rightarrow b} g(x) = L$$

$g(x)$ 不一定连续

The Intermediate Value Property (介值定理):

f 在 $[a, b]$ 上连续, $y_0 \in [f(a), f(b)]$, 则一定有 $c \in [a, b]$, $y_0 = f(c)$

零点存在: f 连续 $f(a) \cdot f(b) < 0$ $[a, b]$

解方程: ① 设函数 ② 找异号 ③ 说连续 ④ 有零点

Continuous Extension:

$$F(x) = \begin{cases} f(x), & x \neq c \\ L, & x = c, \end{cases} \quad (L = \lim_{x \rightarrow c} f(x))$$

At Infinity (∞):

∞ 正 $-\infty$ 负 $\pm\infty$ 讨论

多项式: 分子分母同时除以最高次数 (化为 $\frac{1}{x}$: $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$)

Horizontal Asymptotes: ∞ 或 $-\infty$ 均可 $\begin{cases} 0 \text{ 条} \\ 1 \text{ 条} \\ 2 \text{ 条} \end{cases}$

$$\lim_{x \rightarrow \infty} \sin \frac{1}{x} = \lim_{t \rightarrow 0} \sin t = 0$$

Derivative:

$$\lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h}$$